



Cross-View Geospatial Artificial Intelligence for Disaster Damage Assessment

From Ground-Level Perception to Immersive Decision Support

Yifan Yang, Ph.D. Student, Department of Geography

Committee members:

Dr. Lei Zou (Chair), Dr. Heng Cai, Dr. Andrew Klein, Dr. Zhengzhong Tu

06.24.2026@ Preliminary Exam, O&M 807

“Seeing, describing, and governing disasters — beyond the boundaries of the screen, let the real world become the stage for intelligence.” --Yifan Yang



Outline

- **Background**
- **Current Challenges**
- **Conceptual Framework**
- **Objectives**
- **Case Studies**
- **Research Schedule**

Background

Disaster Damage Assessment Is a Geographic Problem

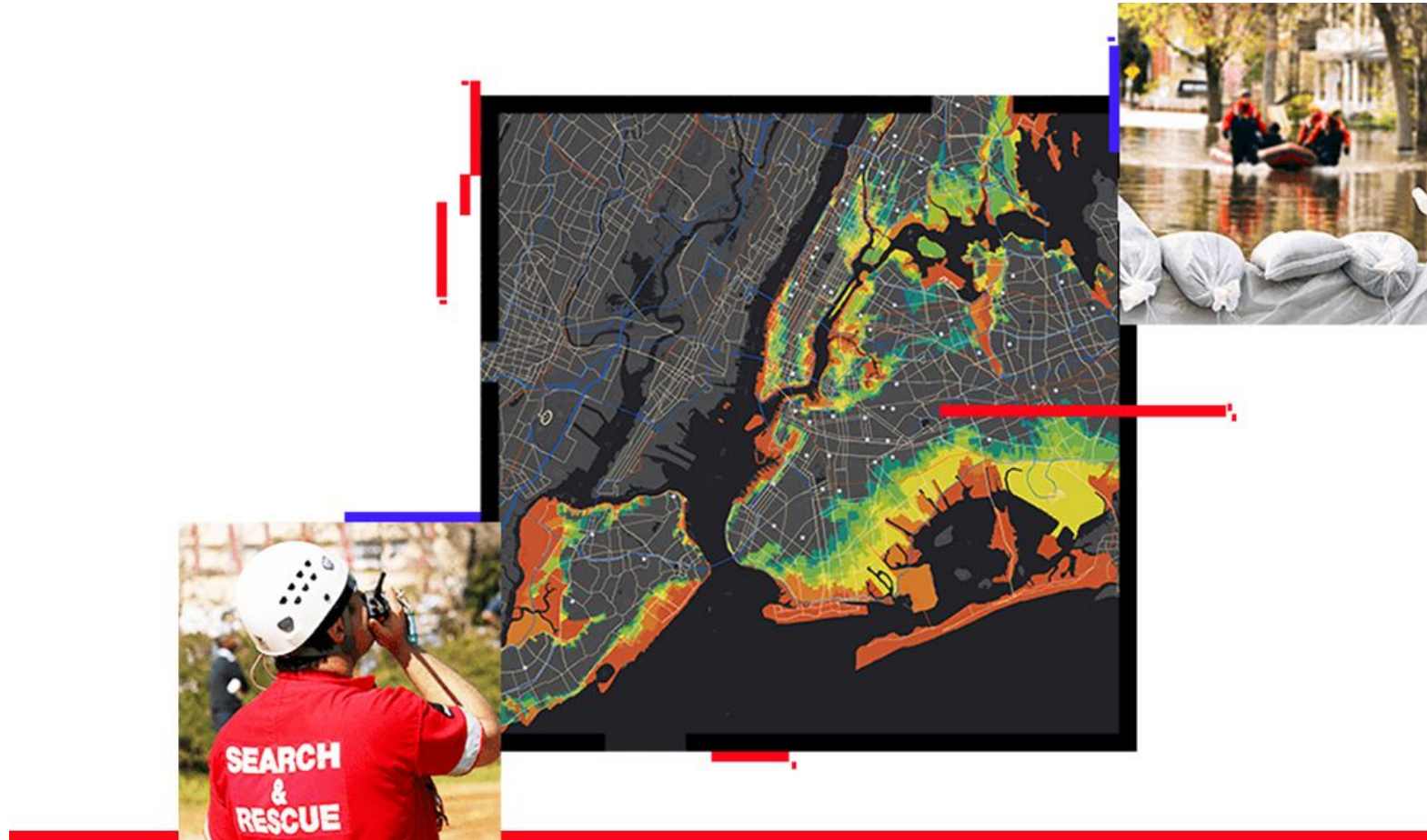


Image courtesy of Esri, taken from its Emergency Management and Damage Assessment solutions webpage (<https://www.esri.com/en-us/industries/emergency-management/solutions/damage-assessment>).

Why Disaster Damage Assessment Matters: Emergency Response

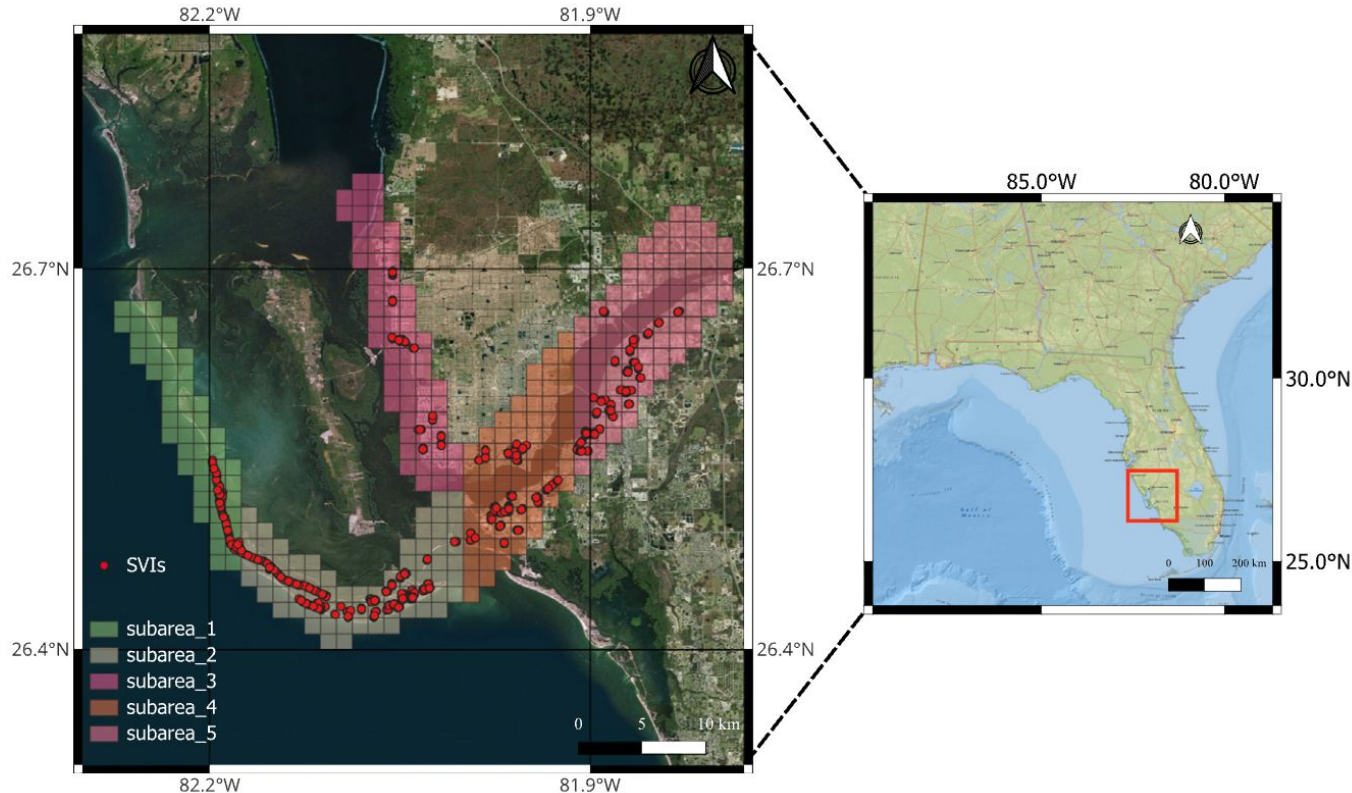


Source: Federal Emergency Management Agency (FEMA). | GAO-25-108598

Emergency Response

- Initial hours after disaster
- Hardest-hit areas
- Rescue and aid prioritization
- Location + speed = lives saved
- Real-time spatial triage

Why Disaster Damage Assessment Matters: Resource allocation



Resource Allocation

- Spatial prioritization
- Areas of urgent need
- Limited personnel, supplies, and funds
- Hardest-hit regions first
- Where should limited resources go?

Li, H., Deuser, F., Yin, W., Luo, X., Walther, P., Mai, G., Huang, W., & Werner, M. (2025). Cross-view geolocalization and disaster mapping with street-view and VHR satellite imagery: A case study of Hurricane IAN. *ISPRS Journal of Photogrammetry and Remote Sensing*, 220, 841-854. <https://doi.org/10.1016/j.isprsjprs.2025.01.003>

Why Disaster Damage Assessment Matters: Recovery & resilience planning

HURRICANE DAMAGE ASSESSMENT

Process showing the steps from safety check to insurance claim submission, including documentation requirements; professional assessment options, and technology tools used for comprehensive evaluation.



Recovery & Resilience Planning

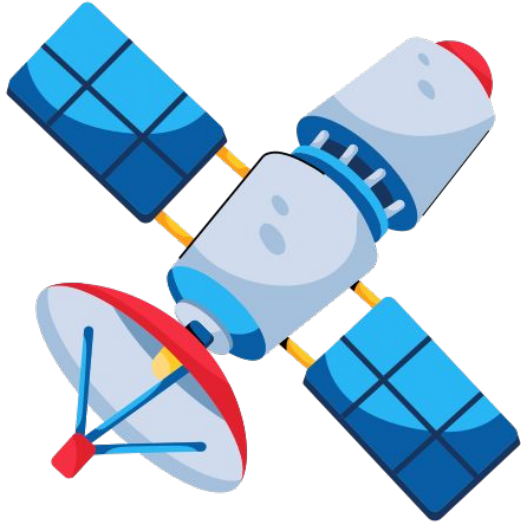
- Property-level damage evidence
- Insurance and recovery claims
- Slow manual inspection
- From assessment to rebuilding
- How can recovery scale across places?

Image Source:
<https://insuranceclaimrecoverysupport.com/hurricane-damage-assessment/>



Background

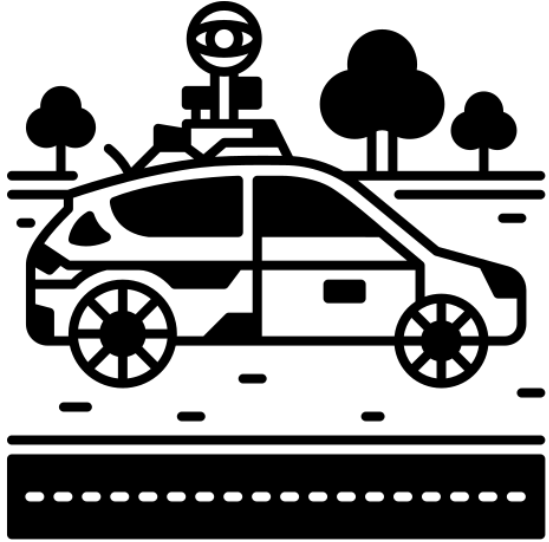
Aerial perspective of disaster impact (Geospatial image data)



Hauptman, L., Mitsova, D., & Briggs, T. R. (2024). Hurricane Ian Damage Assessment Using Aerial Imagery and LiDAR: A Case Study of Estero Island, Florida. *Journal of Marine Science and Engineering*, 12(4), 668. <https://doi.org/10.3390/jmse12040668>

Background

Ground-level perspective of disaster impacts (Geospatial image data)



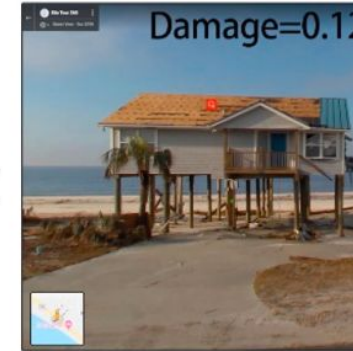
Case A

Ground truth: Damage level=5



Case C

Ground truth: Damage level=4



Case B

Ground truth: Damage level=4



Case D

Ground truth: Damage level=3



Zhai, W., & Peng, Z. (2020). Damage assessment using Google Street View: Evidence from Hurricane Michael in Mexico Beach, Florida. *Applied Geography*, 123, 102252. <https://doi.org/10.1016/j.apgeog.2020.102252>

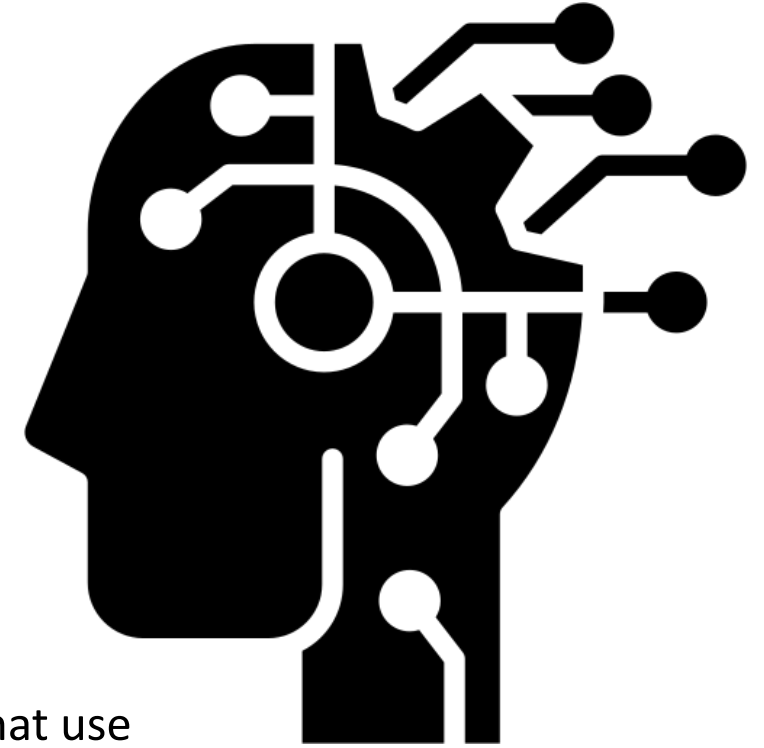
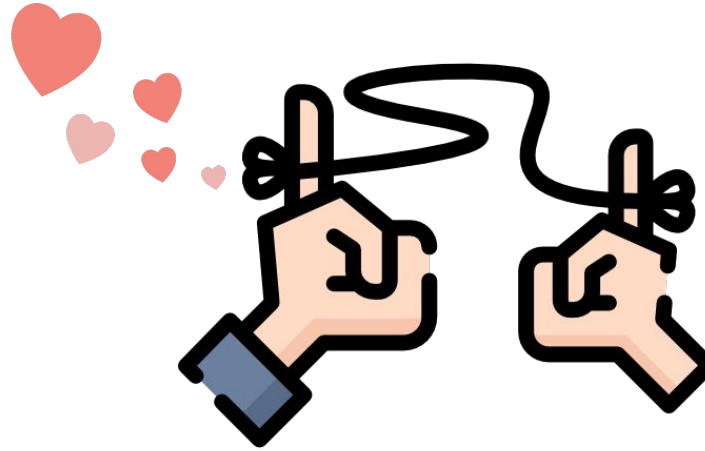
Background

The Engine: Geospatial AI (GeoAI)

Artificial Intelligence



Geography



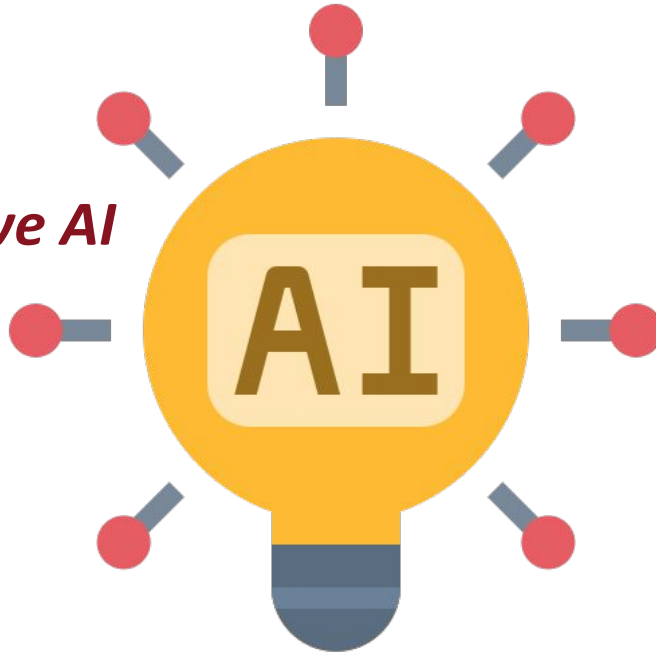
GeoAI refers to **spatially explicit AI** methods that use **geographic data and spatial context** to **create** intelligent geographic information, knowledge, and services.

Janowicz, K., Gao, S., McKenzie, G., Hu, Y., & Bhaduri, B. (2020). GeoAI: spatially explicit artificial intelligence techniques for geographic knowledge discovery and beyond. *International Journal of Geographical Information Science*, 34(4), 625-636.

Background

Two Families of AI — (Discriminative vs. Generative)

Discriminative AI

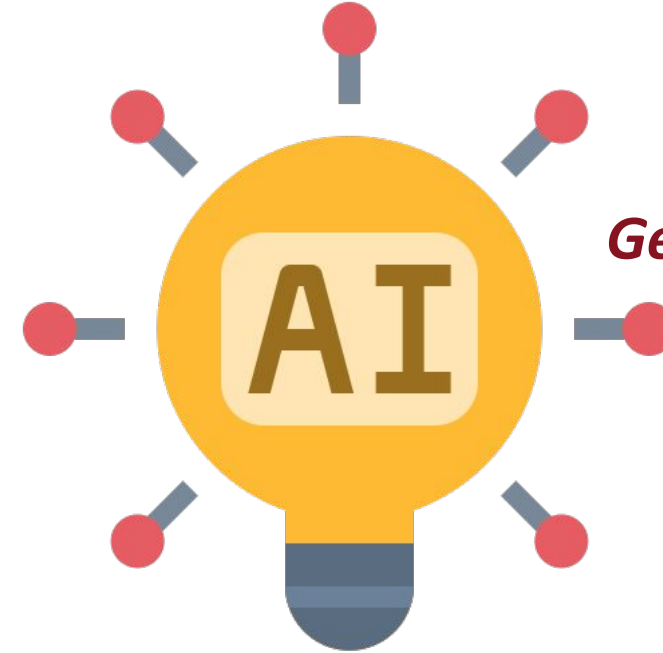


Designed for **closed-ended questions with labeled answers** (such as "what is it" or "to what extent"):

"Is this building damaged?"

"What is the extent of the damage—none/minor/severe/total destruction?"

Generative AI



Open-ended/Constructive Questions ("What is missing? / What if...? / How should it be presented, and why?")

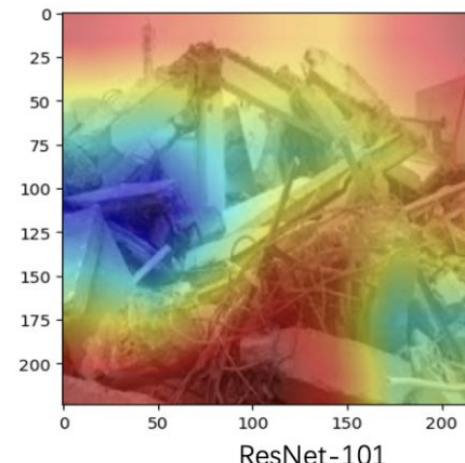
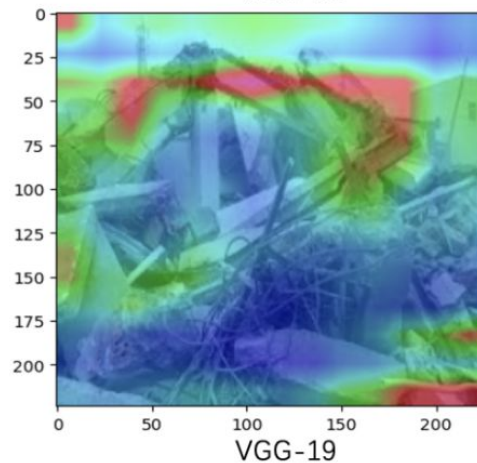
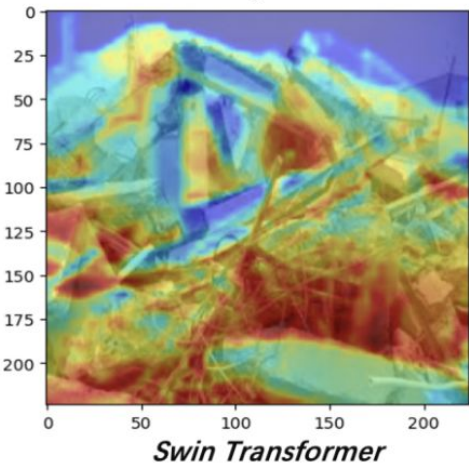
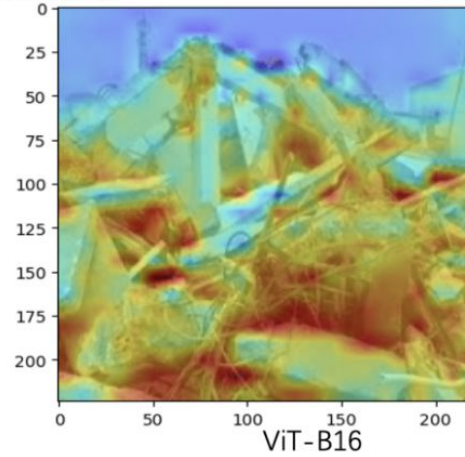
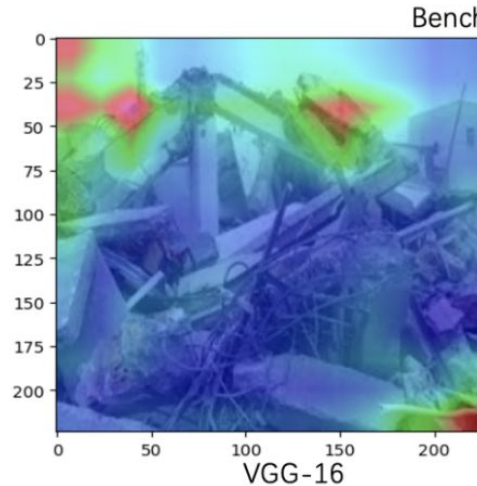
"Why was this location assessed as severely damaged? What actions should rescue personnel take here?"

Background

From GeoAI to Disaster Damage Assessment



Original



GeoAI for Disaster Damage Assessment

- Inputs: satellite, aerial, and street-view imagery
- Tasks: detect damage, classify severity, explain evidence
- Spatial output: hyperlocal damage maps
- Decision goal: support response, recovery, and planning

Zhang, X., Xue, Z., Prevatt, D. O., Bridge, J. A., Xu, S., & Zhao, X. (2026). Post-hurricane building damage assessment using street-view imagery and structured data: a multimodal deep learning approach. *Urban Lifeline*, 4(1), 14.

Current Challenges

1

Place is only partially observed.

This is the problem of seeing: disaster damage is captured only partially across perspective, scale, and time. **See better**

2

Place-specific disasters require rapid assessment without event-specific training.

This is the problem of rapid response: each disaster unfolds in a unique local context, but responders cannot wait for new labels, retraining, or event-specific models. **Respond faster**

3

Place is difficult to simulate, inspect, and communicate.

This is the problem of preparedness and delivery: disaster intelligence should help people understand possible impacts before events and inspect affected environments after events through integrated 2D/3D representations. **Prepare and deliver better**

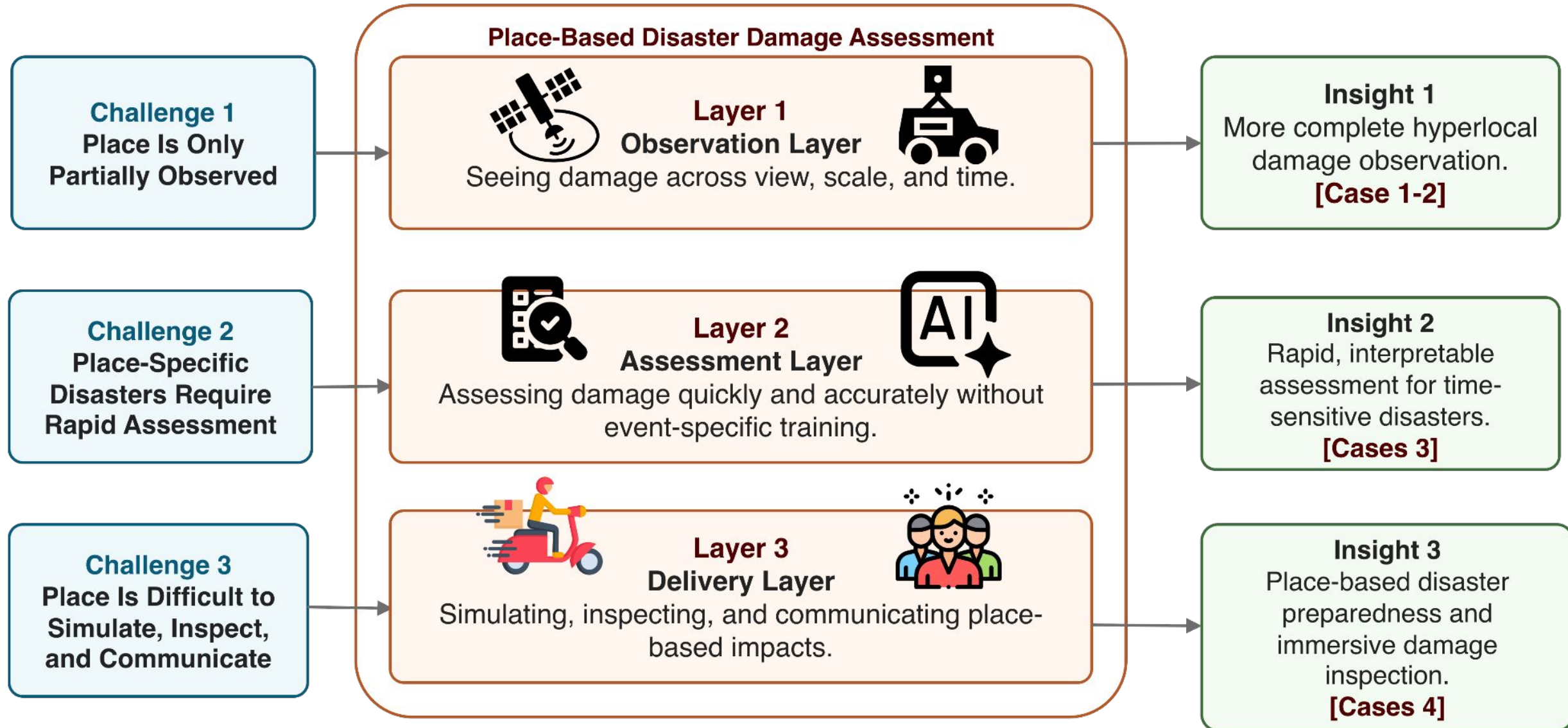


Conceptual Framework

Challenges

Framework Layers

Insights



Objectives

Objective 1 → *addresses Challenge 1* → Case 1,2

Assess disaster damage more completely by observing the same place across multiple views and times, including overhead imagery, ground-level imagery, and pre-disaster baselines. **See better**

Objective 2 → *addresses Challenge 2* → Case 3

Develop rapid and interpretable damage assessment methods that can work with available disaster imagery without new labels, retraining, or event-specific models. **Respond faster**

Objective 3 → *addresses Challenge 3* → Case 4

Create place-based 2D/3D representations that support disaster preparedness, post-event inspection, and communication for responders, residents, and decision makers. **Prepare and deliver better**

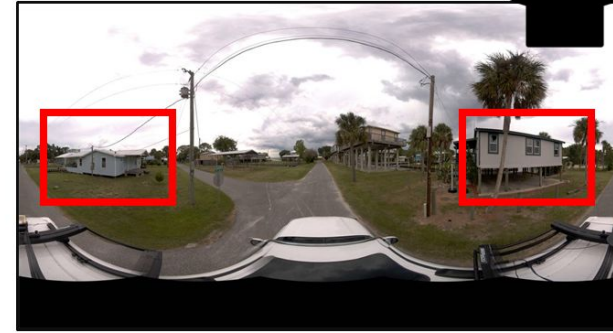
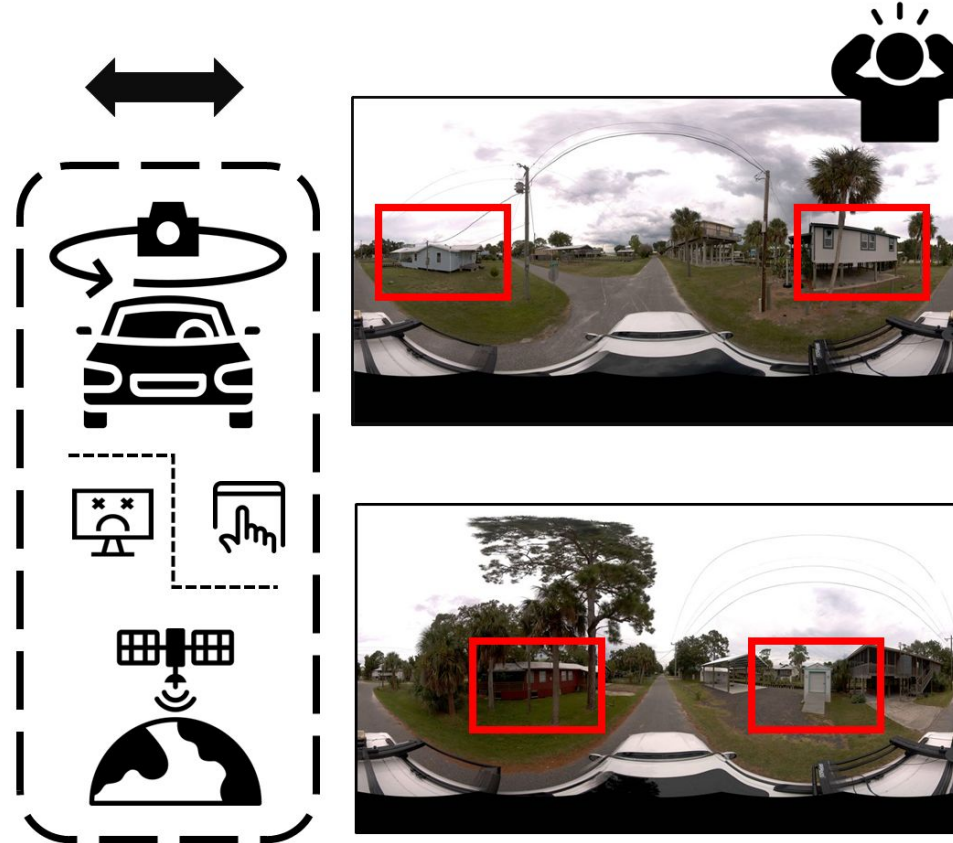


Case 1:

Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models
(Published, CEUS 2025)



After Hurricane Milton

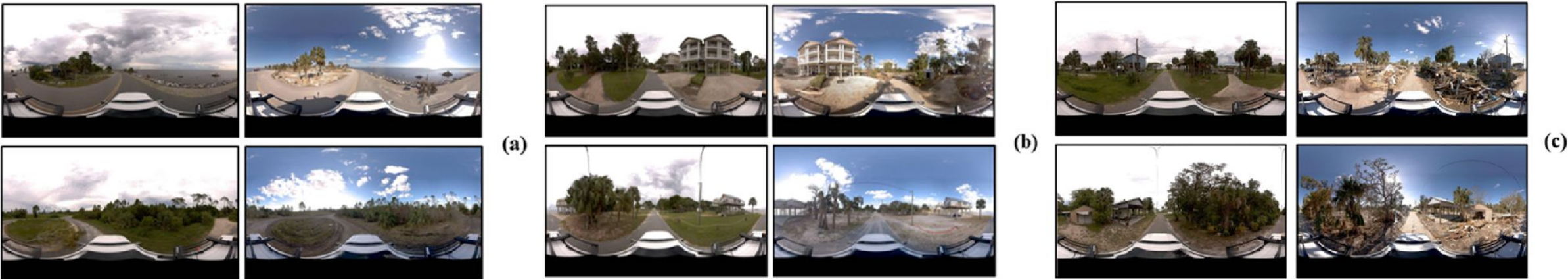
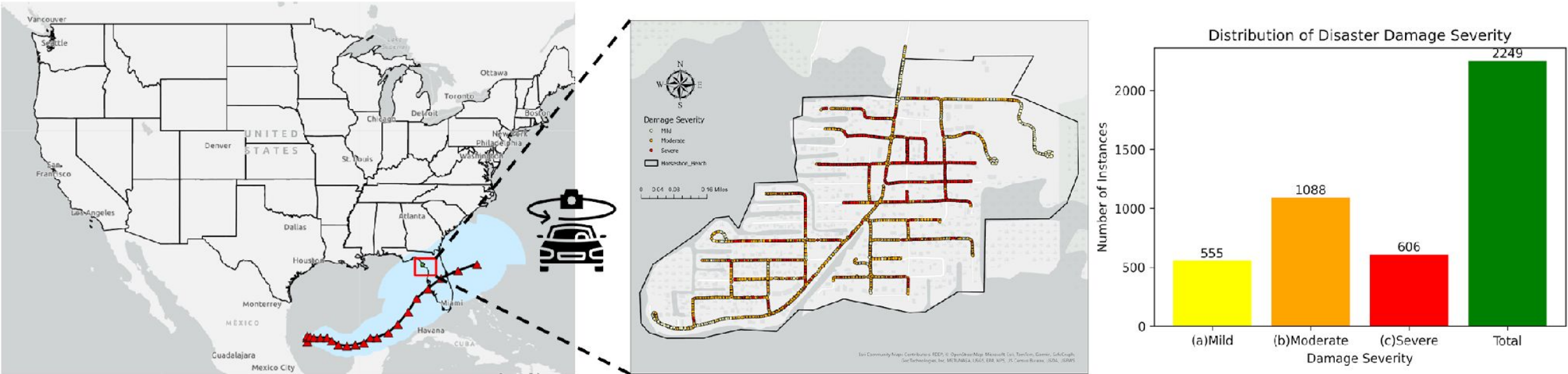


Before Hurricane Milton

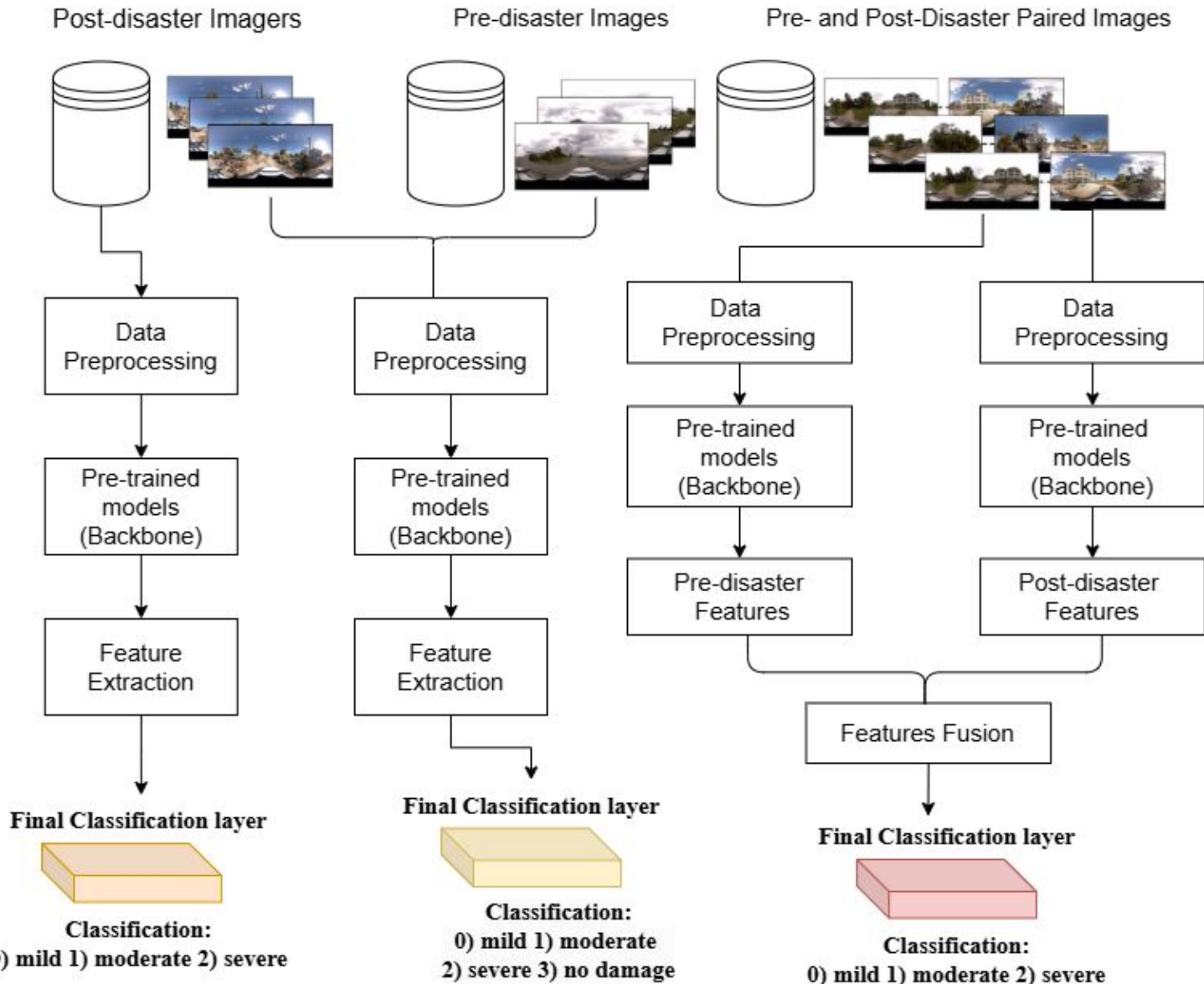
Research questions:

- (1) Can **pre-disaster street-view imagery** help us classify disaster damage **more accurately**?
- (2) How can we **integrate pre- and post-disaster** street-view imagery to enable rapid, hyperlocal damage assessment?

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (study area and dataset)



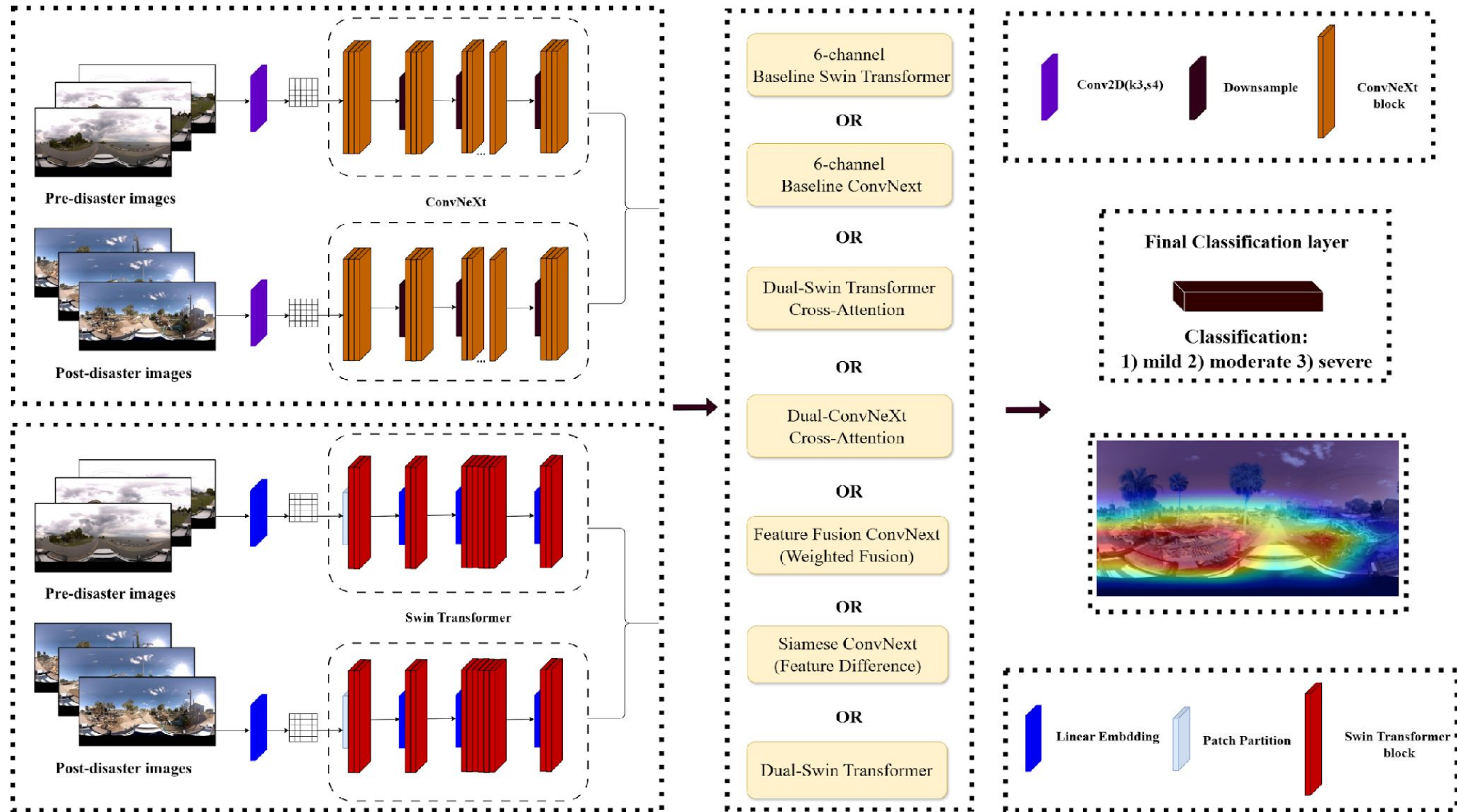
Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (methology)



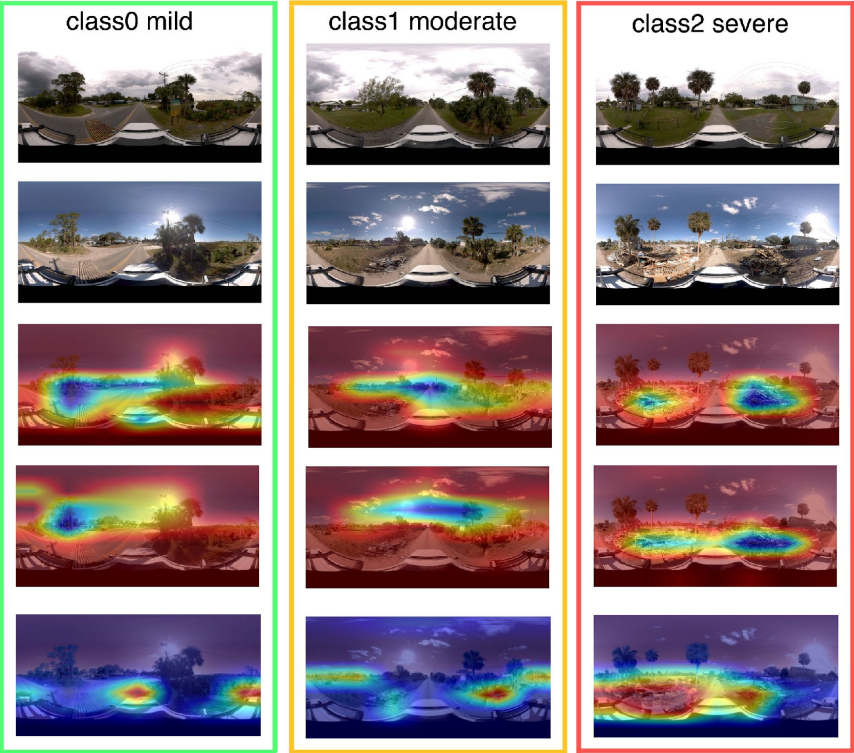
Method

- Post-disaster image baseline
- Pre-disaster image baseline
- Pre/post paired image learning
- Fine-tuned Swin Transformer and ConvNeXt
- Dual-channel feature fusion
- Grad-CAM explainability

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models



Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models



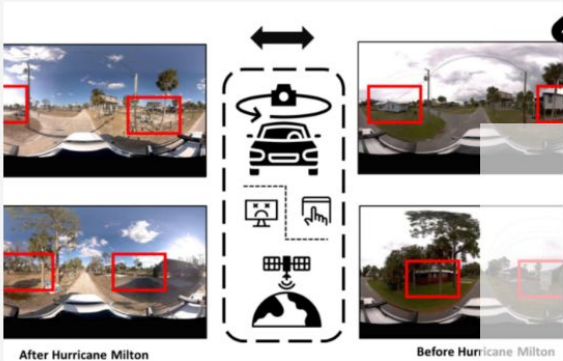
	Model name	Accuracy	P	R	F1	Class-Specific Accuracy		
						0-mild	1-moderate	2-severe
(A)	Baseline-ConvNext	72.80%	0.7315	0.7280	0.7256	56.91%	79.12%	75.54%
	Baseline-Swin Transformer	66.14%	0.6889	0.6614	0.6481	39.02%	84.34%	57.55%
(B)	Dual-ConvNeXt + Cross-Attention	76.91%	0.7704	0.7691	0.7672	71.08%	72.36%	91.37%
(C)	Dual-Swin Transformer + Cross-Attention	76.91%	0.7970	0.7691	0.7615	51.47%	91.97%	74.60%
(D)	Dual-Swin Transformer	74.17%	0.7420	0.7417	0.7408	62.69%	75.45%	81.33%
(E)	Feature-fusion ConvNext	77.11%	0.7751	0.7710	0.7678	59.39%	82.72%	84.44%
	Siamese ConvNeXt	75.54%	0.7878	0.7554	0.7498	53.28%	90.79%	71.11%

Key result: Accuracy improved from 66.14% (Swin baseline) to 77.11% (dual-channel Feature-Fusion ConvNeXt); Grad-CAM confirms pre-disaster images help focus on true pre/post changes.

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (Published, CEUS 2025)

Mosaic

ABOUT ▾PRODUCTS ▾OFFERINGS ▾CASE STUDIESINSIGHTS ▾✉



How Texas A&M is Pioneering Damage Assessment Using Mosaic Imagery from Hurricane Milton

CAROLINE APRIL 15, 2026 FEATURED

READING TIME: 7 MINUTES

<https://www.mosaic51.com/featured/how-texas-am-is-pioneering-damage-assessment-using-mosaic-imagery-from-hurricane-milton/>

Yang, Y., Zou, L., Zhou, B., Li, D., Lin, B., Abedin, J., & Yang, M. (2025). Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models. Computers, Environment and Urban Systems, 121, 102335.

ELSEVIER

Contents lists available at ScienceDirect

Computers, Environment and Urban Systems

journal homepage: www.elsevier.com/locate/ceus

COMPUTERS
ENVIRONMENT
AND
URBAN SYSTEMS

Check for updates

Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models

Yifan Yang^a, Lei Zou^{a,*}, Bing Zhou^{b,c}, Daoyang Li^d, Binbin Lin^a, Joynal Abedin^a, Mingzheng Yang^a

^a Department of Geography, Texas A&M University, College Station, USA

^b Department of Geography, Pennsylvania State University, University Park, USA

^c Department of Geography and Sustainability, University of Tennessee, Knoxville, USA

^d Spatial Sciences Institute, University of Southern California, Los Angeles, USA

ARTICLE INFO

Keywords:
Disaster resilience
Street-view imagery
Dual-channel neural network
Pre-trained vision model
Damage estimation

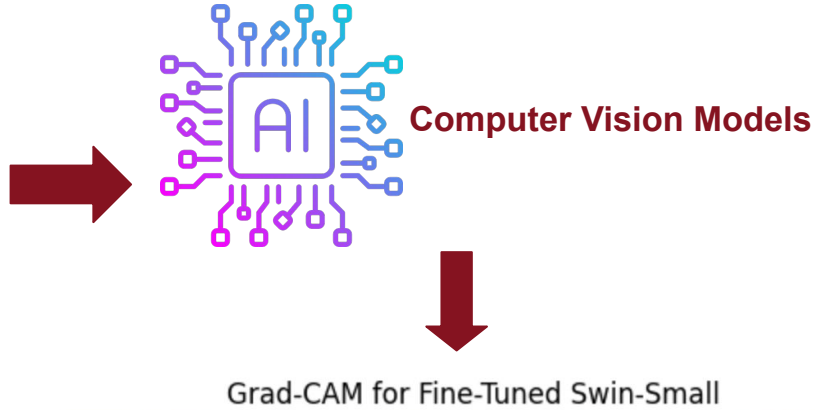
ABSTRACT

Street-view images offer unique advantages for disaster damage estimation as they capture impacts from a visual perspective and provide detailed, on-the-ground insights. Despite several investigations attempting to analyze street-view images for damage estimation, they mainly focus on using post-disaster images. The potential of time-series street-view images remains underexplored. Pre-disaster images provide valuable benchmarks for accurate damage estimations at building and street levels. These images could also aid annotators in objectively labeling post-disaster impacts, improving the reliability of labeled data sets for model training, and potentially enhancing the model performance in damage evaluation. The goal of this study is to estimate hyperlocal, on-the-ground disaster damages using bi-temporal street-view images and advanced pre-trained vision models. Street-view images before and after 2024 Hurricane Milton in Horseshoe Beach, Florida, were collected for experiments. The objectives are: (1) to assess the performance gains of incorporating pre-disaster street-view images as a no-damage category in fine-tuning pre-trained models, including Swin Transformer and ConvNeXt, for damage level classification; (2) to design and evaluate a dual-channel algorithm that reads pair-wise pre- and post-disaster street-view images for hyperlocal damage assessment. The results indicate that incorporating pre-disaster street-view images and employing a dual-channel processing framework can significantly enhance damage assessment accuracy. The accuracy improves from 66.14 % with the Swin Transformer baseline to 77.11 % with the dual-channel Feature-Fusion ConvNeXt model. Gradient-weighted Class Activation Mapping (Grad-CAM) shows that incorporating pre-disaster images improves the pre-trained vision model's capacity to focus on major changes between pre- and post-disaster images, thus enhancing model performance. This research evaluates the technical solutions and challenges of assessing disaster damages at hyperlocal spatial resolutions with bi-temporal street-view images, providing valuable insights to support effective decision-making in disaster management and resilience planning.

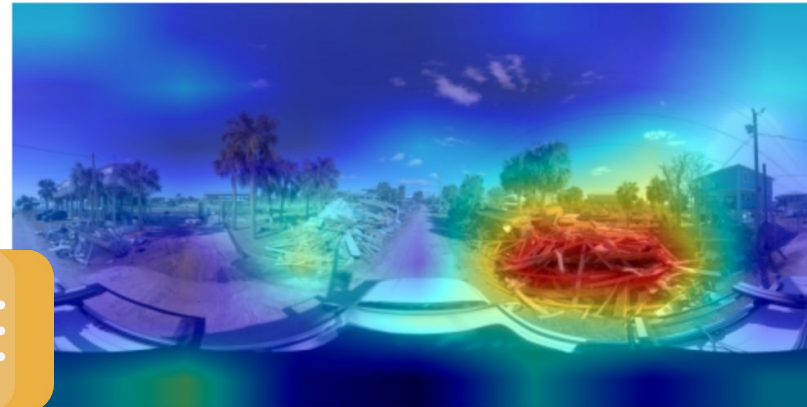
Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery (*Submitted.*)



Post-Hurricane
Street-View Imagery



Severe



Motivation

- Accuracy alone is not enough
- Reliability matters for disaster response
- Overconfident wrong predictions
- Risk of misleading prioritization

Overconfidence=models may give very confident predictions even when they are **wrong**

Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery

class0 mild





No buildings in the area were damaged by the hurricane, and the trees appear to have sustained only mild damage. This is truly a relief. The vegetation is lush, the trees are tall, and the damage from the hurricane was minimal. I'd love to live here now, as it's only slightly damaged and looks very livable.



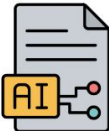
Sagging power lines are visible alongside a clear, dry road. Many trees show defoliation and some branch damage, but no large fallen trees or building debris are present. The overall scene suggests mild environmental impact, with infrastructure largely intact.

class1 moderate





The damage level is moderate, with loose building materials and a small amount of debris suggesting recent construction or minor disrepair. Despite the somewhat cluttered appearance of scattered items, the water features and elevated residences make this area appear to be an attractive residential area.



Scattered wooden debris and loose utility lines on a power pole indicate moderate damage along this coastal dirt road. While buildings appear structurally sound, the presence of scattered planks and beams, alongside the compromised power infrastructure, suggests the area experienced some impact, but not widespread destruction or flooding.

class2 severe





The atmosphere of the image is chaotic, with massive piles of debris covering large parts of the area, giving a strong sense of destruction. The damage appears extensive, with shattered building materials strewn all over, although some elevated houses remain intact in the background.



The street is heavily impacted by extensive building debris, forming large piles along both sides, almost completely blocking passage. This significant accumulation of scattered materials and collapsed fragments suggests severe damage to structures in the vicinity, though visible infrastructure like power lines appears mostly intact.

RQ1: How **reliable** are image-only, text-only, and image-text models for street-view disaster damage assessment?

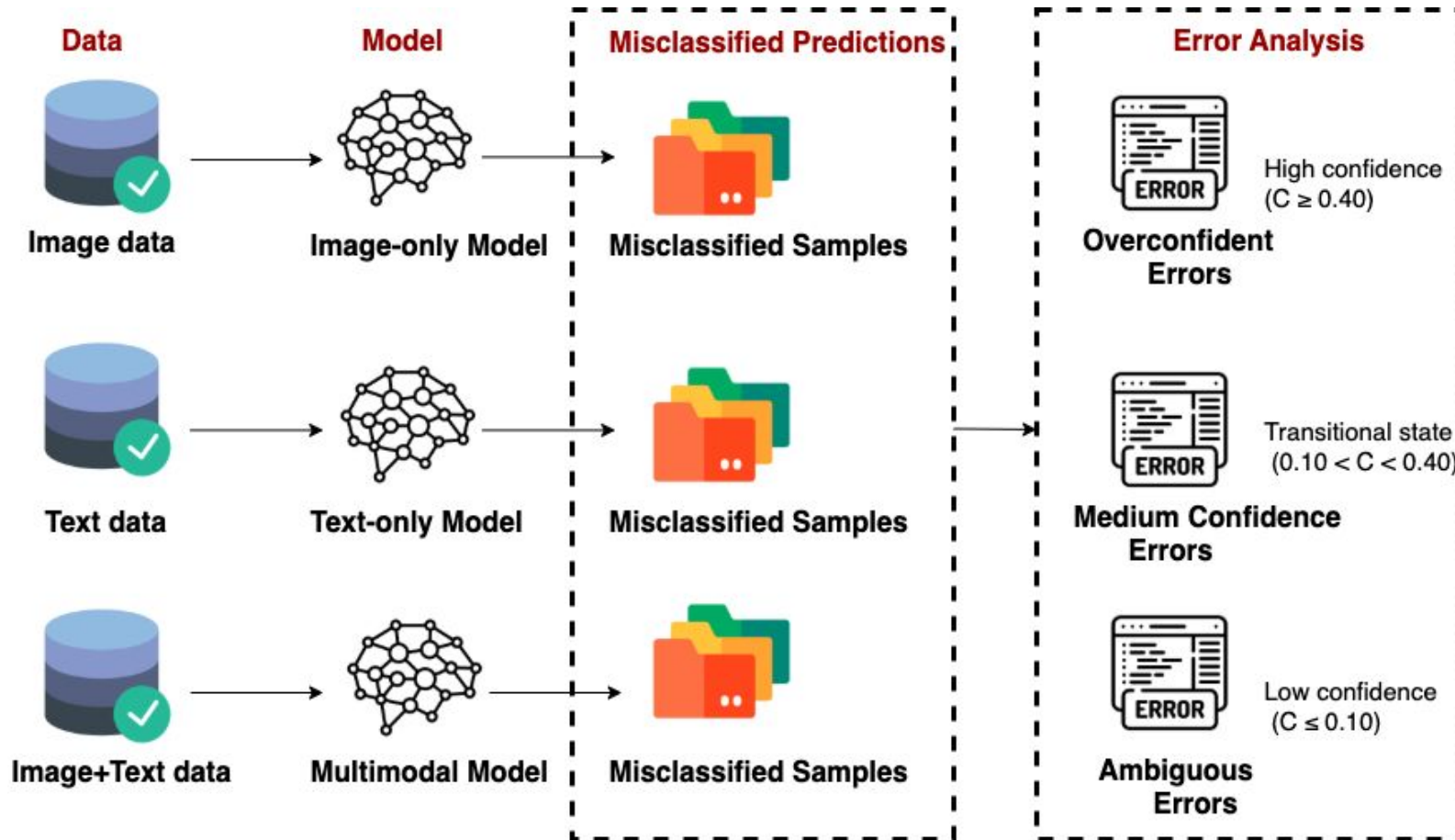
RQ2: Can LLM-generated image descriptions **support reliable damage assessment** when human annotations are limited?

 **EAR Lab**
Explore & Resolve

Yifan Yang | Ph.d. Student@Texas A&M University | Email: yyang295@tamu.edu | Lab Website: <https://www.geoeearlab.com>

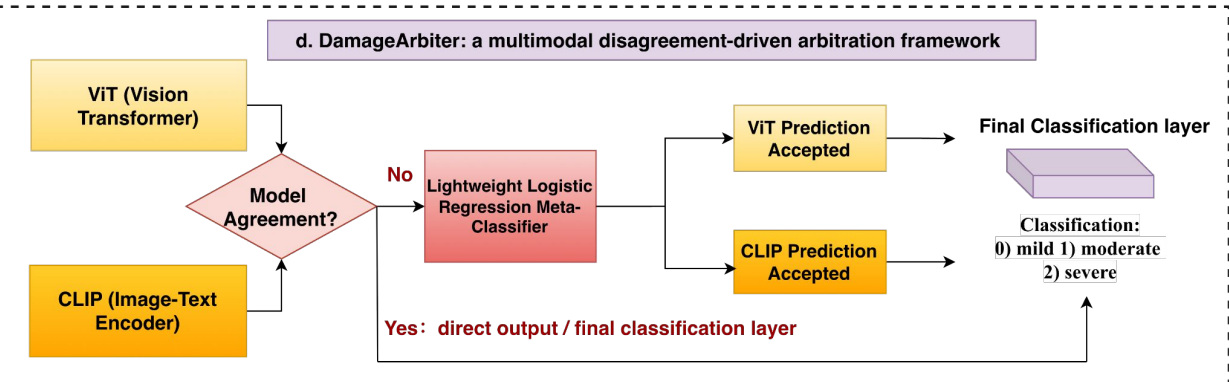
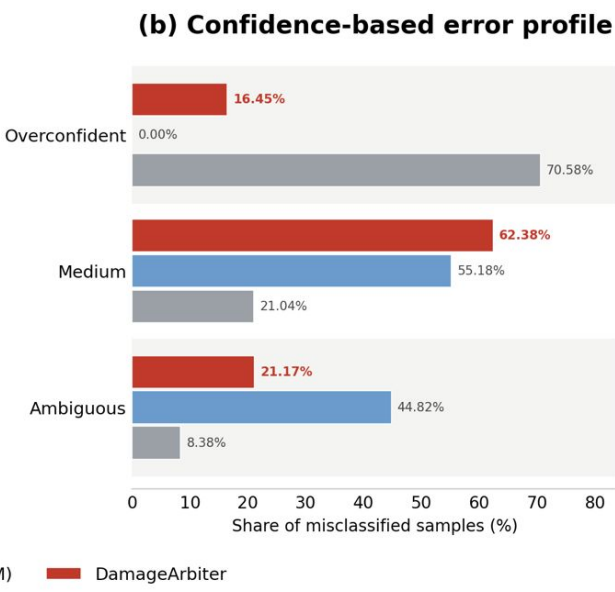
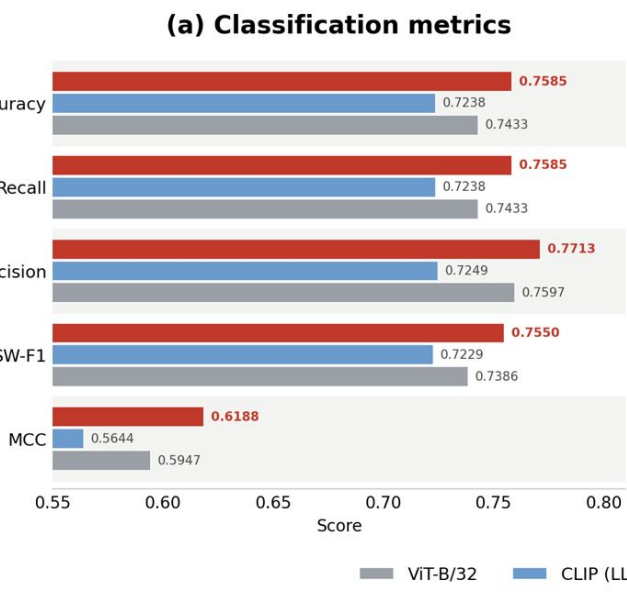
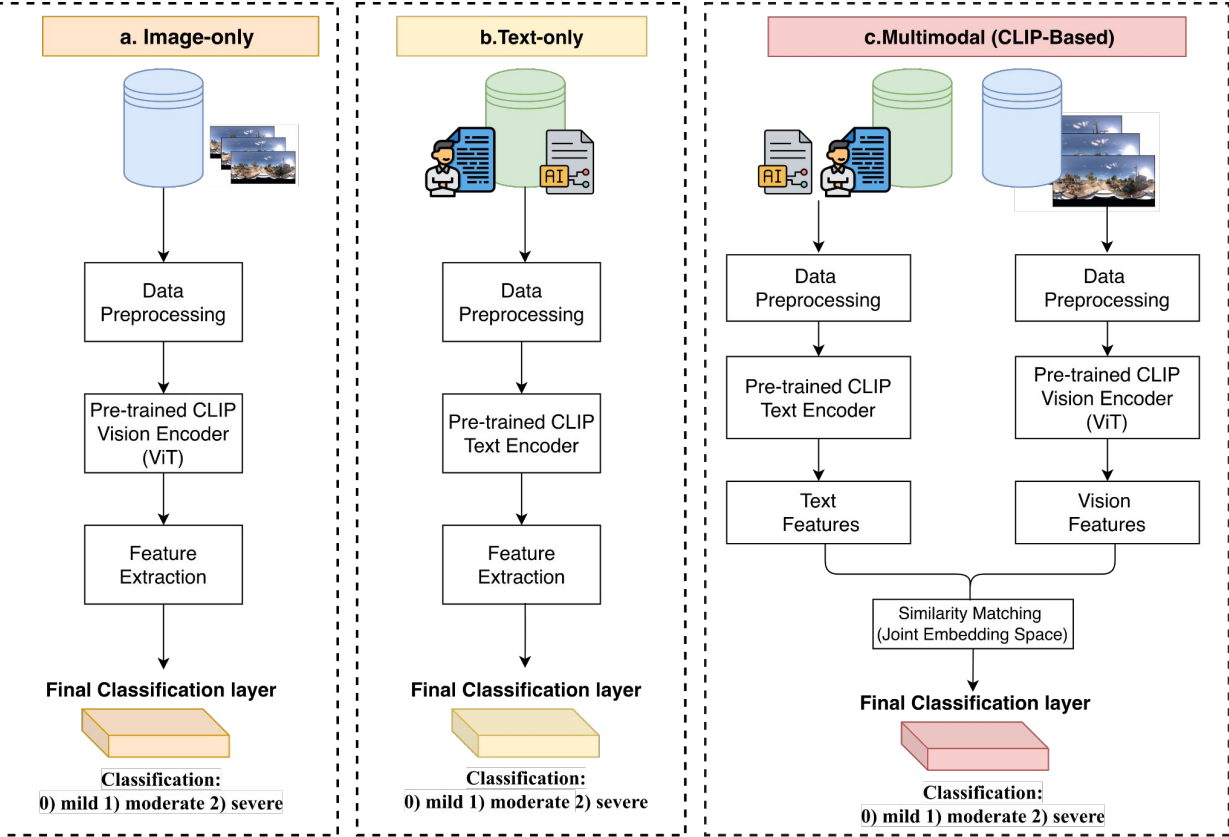
22

Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery

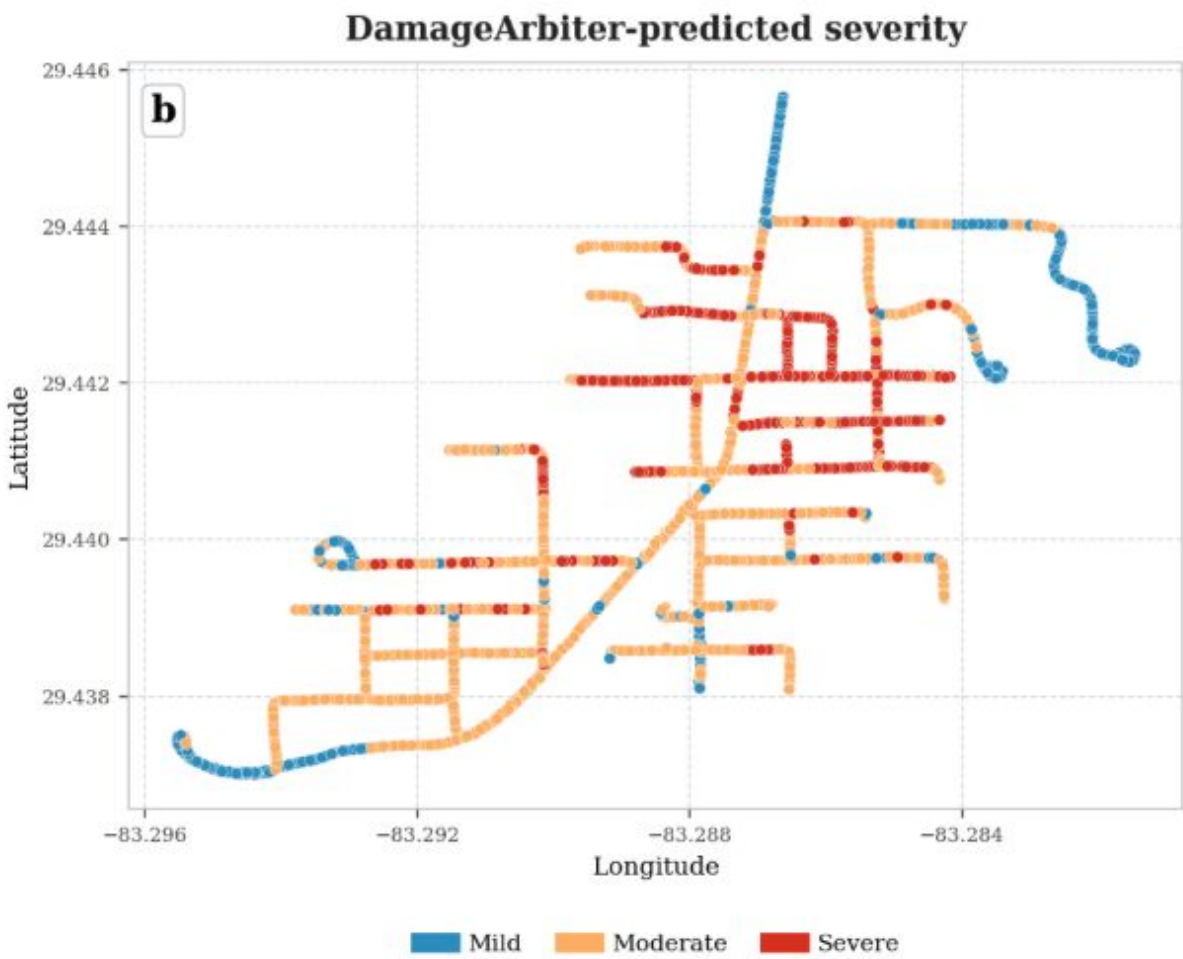
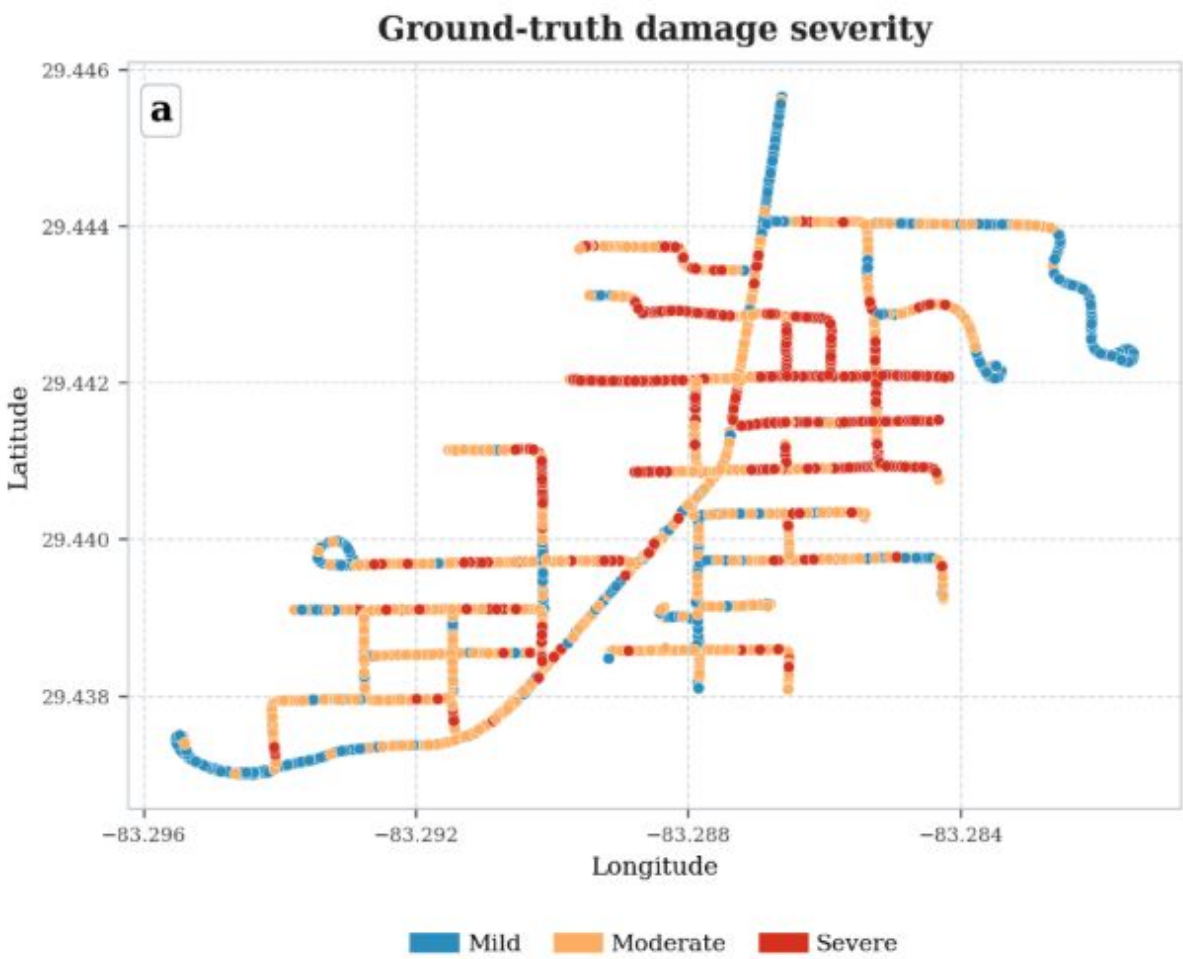


Overconfidence=models may give very confident predictions even when they are **wrong**

Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery



Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery



Case 3:

RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery (*Under revision*)



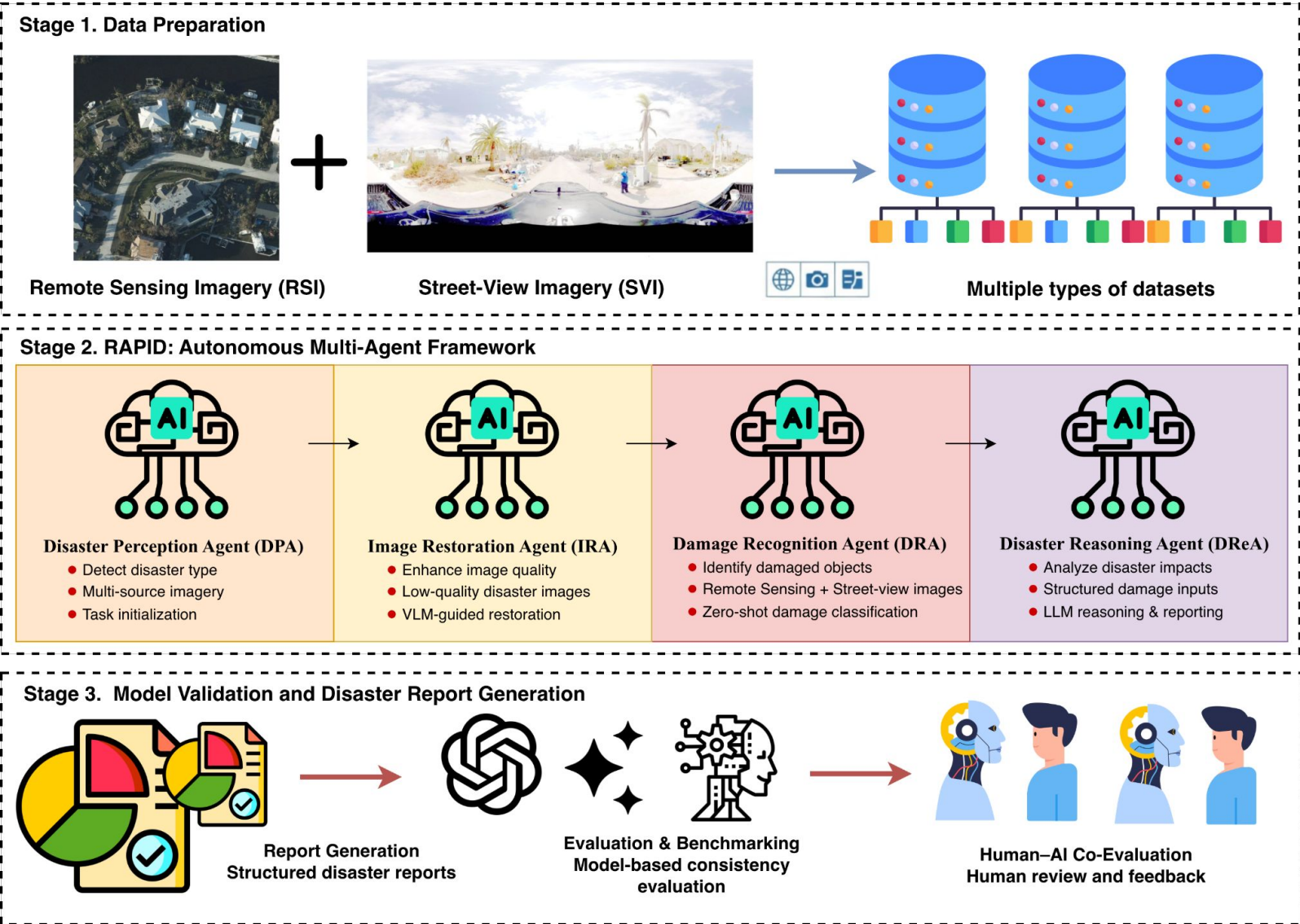
Research Question

Can RAPID support:

- Cross-view assessment?
- Different regions and hazards?
- Rapid zero-shot reasoning?
- Interpretable damage reports?



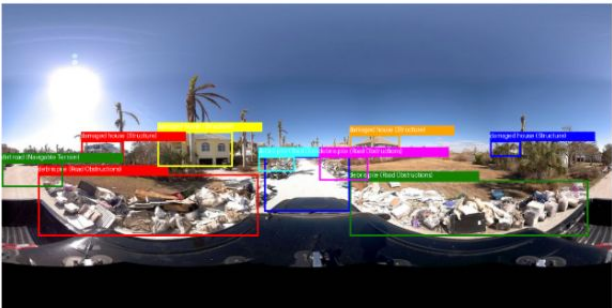
Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery



Method:

- Cross-view imagery input
- Zero-shot multi-agent assessment
- Structured report generation

Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

Input Images	Model Responses	Gemini-3-Pro	Gemini-2.5-Pro	ChatGPT-5.1
 	Type of disaster	hurricane	hurricane	hurricane
	Image restoration	No	No	No
	Disaster severity classification	Severe	Severe	severe
	Voted candidate	SVI and RSI	SVI and RSI	SVI and RSI
	Confidence	0.85	0.95	0.96
	Object recognition	Debris, Fallen Tree	Debris, Fallen Tree, Damaged Building	Debris, Fallen Tree, Damaged Building
 Using Gemini-3-Pro for object recognition	The Street View Image reveals massive piles of debris (furniture, appliances, drywall) lining the road, indicating that the adjacent homes were gutted by storm surge, which constitutes 'significant building debris' and major economic destruction. While the roads are not currently 'fully flooded' (they are covered in surge sand) and trees are mostly standing (though stripped), the sheer volume of property loss and the saturated ground visible in the Remote Sensing image align the overall impact with the 'Severe' classification. Reasoning for Gemini-3-Pro-Disaster Assessment			
	Recovery efforts must prioritize the rapid removal of the substantial debris (furniture, appliances, drywall) currently lining the streets to fully restore logistical access. Given that homes were 'gutted' by storm surge, reconstruction requires rigorous structural inspections and professional mold remediation before habitation is permitted. Infrastructure teams should focus on clearing surge sand from roadways and assessing subsurface utility stability due to the 'saturated ground.' Long-term planning should enforce elevated construction standards to mitigate future surge risks. Reasoning for Gemini-3-Pro-Disaster Recovery / Reconstruction Suggestion			

Case 4: From Static Disaster Maps to Immersive Place-Based Damage Visualization
(In the planning stage)



Research Question

- How can ground + overhead images create immersive 3D damage representations?

Planned Output

- Interactive 3D disaster visualization prototype
- Linked damage severity + visual evidence
- User study: 2D maps vs. immersive 3D assessment

Generated using GPT-image, as planned.

Research Schedule

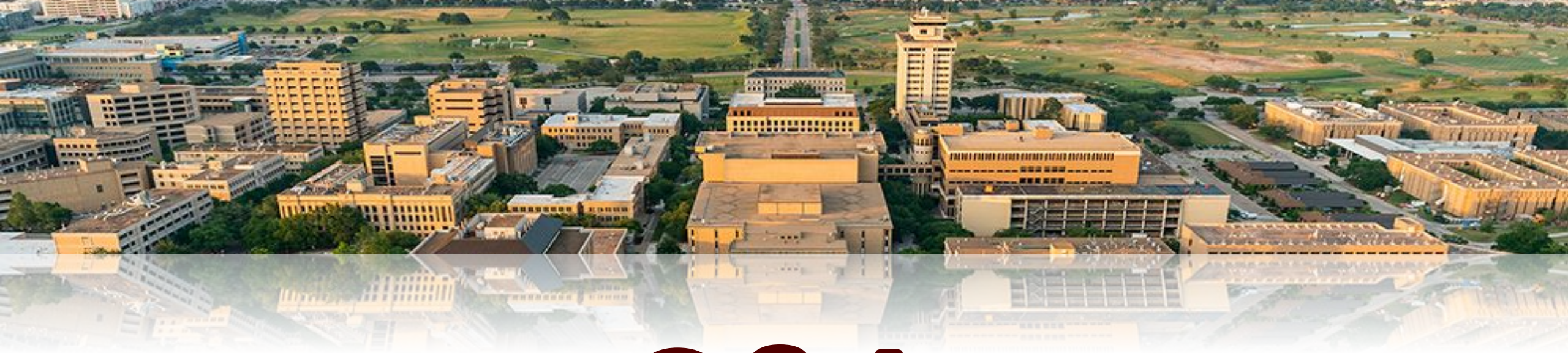


Workstream	Fall 2024	Spring 2025	Fall 2025	Spring 2026	Fall 2026	Spring 2027	Fall 2027	Spring 2028	Status
Case 1 Observation: bi-temporal street-view damage assessment	Published								Done
Case 2 Assessment: DamageArbiter reliability and arbitration			Submitted						In Progress
Case 3 Assessment: RAPID cross-view disaster reports				Revision					In Progress
Case 4 Delivery: 3D place-based disaster visualization					Design + user study				Planning
Prelim Exam Program milestone				Exam					In Progress
Proposal Defense Program milestone						Defense			Not Started
Dissertation Writing, defense, and deposit							Writing + defense		Not Started

Legend: Done | In Progress | Planning | Not Started

This dissertation builds cross-view geospatial artificial intelligence that lets us see, describe, and govern disasters.





Q&A

Welcome any comments/questions

yyang295@tamu.edu

